NAME: Rajvardhan Patil
ROLL NO.: 2401081
BATCH: T21
BRANCH: AI&DS

EXPERIMENT – 1

Aim: Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory:

Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

Commands:

git init : Initializes a new Git repository in the current working directory. git clone

<repository-url> : Creates a local copy of an existing remote GitHub repository.

git status : Displays the current state of the repository, including staged, modified, and untracked files.

git add <file> : Adds a specific file to the staging area before committing. git

add . : Stages all newly created and modified files in the current directory.

git commit -m "message" : Records the staged changes in the local repository with a descriptive commit message.

git log : Shows the complete history of commits made in the repository. git log --oneline :

Displays a concise, one-line summary of each commit. git diff : Shows the differences

between the working directory and the most recent commit. git branch : Lists all available

local branches and indicates the active branch. git branch <branch-name> : Creates a new

branch without switching to it. git checkout <branch-name> : Switches from the current

branch to the specified branch.

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git checkout -b <branch-name> : Creates a new branch and immediately switches to it. git

switch <branch-name> : Switches to an existing branch using the modern Git command.

git merge <branch-name> : Merges the specified branch into the currently active branch.

git remote -v : Displays the remote repositories linked to the local repository.

git remote add origin <repository-url> : Connects the local repository to a remote GitHub repository.

git push origin <branch-name> : Uploads local commits from the specified branch to the remote repository.

git pull origin <branch-name> : Fetches and merges the latest changes from the remote repository into the current branch.

git fetch : Downloads updates from the remote repository without merging them into the current branch.

git reset --hard HEAD : Removes all uncommitted changes and restores the working directory to the latest commit.

git stash : Temporarily saves uncommitted changes without creating a commit.

git stash pop : Restores the most recently stashed changes and removes them from the stash list.

git rm <file> : Deletes a file from both the working directory and the Git repository.

git config --global user.name "Your Name" : Sets the global username that Git uses for commits.

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```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\AI&DS> git --version
git version 2.55.0.windows.3
PS C:\Users\AI&DS> git config --global user.name "RajvardhanPatil87"
PS C:\Users\AI&DS> git config --global user.email "rajpatil8787@gmail.com"
PS C:\Users\AI&DS> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
```

```
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=RajvardhanPatil87
user.email=rajpatil8787@gmail.com
PS C:\Users\AI&DS> mkdir GitDemo2


    Directory: C:\Users\AI&DS
```

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```
PS C:\Users\AI&DS> cd GitDemo2
PS C:\Users\AI&DS\GitDemo2> git init
Initialized empty Git repository in C:/Users/AI&DS/GitDemo2/.git/
PS C:\Users\AI&DS\GitDemo2> echo "Hello World" > index.html
PS C:\Users\AI&DS\GitDemo2> echo "body{}" > style.css
PS C:\Users\AI&DS\GitDemo2> 'console.log("Hello");' > app.js
PS C:\Users\AI&DS\GitDemo2> dir


    Directory: C:\Users\AI&DS\GitDemo2


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        23-07-2026     11:50             48 app.js
-a----        23-07-2026     11:50             28 index.html
-a----        23-07-2026     11:50             18 style.css


PS C:\Users\AI&DS\GitDemo2> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.js
        index.html
        style.css
```

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```
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\AI&DS\GitDemo2> git add .
PS C:\Users\AI&DS\GitDemo2> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   app.js
        new file:   index.html
        new file:   style.css

PS C:\Users\AI&DS\GitDemo2> git commit -m "Initial Commit"
[master (root-commit) 889bd51] Initial Commit
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 app.js
 create mode 100644 index.html
 create mode 100644 style.css
PS C:\Users\AI&DS\GitDemo2> git remote add origin https://github.com/RajvardhanPatil87/GitDemo1.git
PS C:\Users\AI&DS\GitDemo2> git remote -v
origin  https://github.com/RajvardhanPatil87/GitDemo1.git (fetch)
origin  https://github.com/RajvardhanPatil87/GitDemo1.git (push)
PS C:\Users\AI&DS\GitDemo2> git branch
* master
```

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```
 create mode 100644 style.css
PS C:\Users\AI&DS\GitDemo2> git remote add origin https://github.com/RajvardhanPatil87/GitDemo1.git
PS C:\Users\AI&DS\GitDemo2> git remote -v
origin  https://github.com/RajvardhanPatil87/GitDemo1.git (fetch)
origin  https://github.com/RajvardhanPatil87/GitDemo1.git (push)
PS C:\Users\AI&DS\GitDemo2> git branch
* master
PS C:\Users\AI&DS\GitDemo2> git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 391 bytes | 391.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/RajvardhanPatil87/GitDemo1.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
PS C:\Users\AI&DS\GitDemo2>
```

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CONCLUSION

The experiment demonstrated the use of essential Git version control commands for repository management, tracking changes, branch handling, and collaboration with GitHub. It provided practical knowledge of managing and maintaining software projects efficiently using Git.